Abstract

Public land-cover products are increasingly used where authoritative local time series are unavailable, but product choice can alter both apparent urban change and model rankings. We present an auditable protocol for constrained spatial allocation using Geospatial Kernel, the algorithmic core of a Geospatial World Model, as the focal execution layer. Two annual 10-m products for Abu Dhabi city, Google Dynamic World (2017–2024) and an ArcGIS-served Impact Observatory/Microsoft/Esri series (2017–2025), were harmonized to the same 100-m, six-class modelling contract. Leakage-controlled expanding-window tests compared a FLUS-style ANN–CA, Geospatial Kernel, GeoFM-LDN, persistence and random minimum-change allocation using three seeds and 1,000 spatial-block bootstrap resamples. The target year supplied oracle class totals and evaluation labels only. In 2024, the ArcGIS-served series mapped 335.57 km² as built versus 155.98 km² in Dynamic World; built-class intersection over union was 0.449. Kernel ranked first in all four Dynamic World targets from 2021 to 2024, but in only two of four matched ArcGIS-served targets; FLUS-style led in 2021 and GeoFM-LDN in 2022. Kernel's ArcGIS-minus-Dynamic World strict change Figure of Merit ranged from -0.244 to +0.205. Yet the product-specific 2031 planning frontiers retained FLUS-style and Kernel candidates in all three scenarios. The results support an inspectable proposal–projection–writeback boundary while showing that comparative skill is label-product dependent. Neither product is authoritative local land-use truth, model outputs remain 100 m, and the 2026–2031 maps are conditional stress tests rather than official forecasts.